

Problems on Pipelined datapath and hazard

Question 01

Ques 1 - 4 detect if there is any data hazard in the code sequences or not

1. `ld x5, 20(x6)`
`addi X20, x5, 45`
2. `addi X20, x5, 45`
`sd x20, 20(x6)`
3. `addi X20, x5, 45`
`sub x21, x20, x5`
4. `ld x5, 20(x6)`
`addi X21, x22, 45`
`addi X23, x22, 45`
`addi X24, x22, 45`
`addi X25, x22, 45`
`addi X20, x5, 45`

Question 02

1. `Add x5, x6, x20`
 2. `addi x22, x6, 75`
 3. `addi x23, x6, 778`
 4. `Lw x6, 24(x6)`
 5. `Sub x6, x5, x5`
- Clock period = 500 ms
- a. Calculate the total amount of time required to execute the instructions using pipelined datapath.
 - b. Is there occurring any data hazard as x5 is the destination register in the first instruction and source register in the fifth instruction? Describe